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Web applications

Web applications are network applications based on a client/server model

Client side:

- User agents, i.e., browsers or software agents, are the client component of the network application running on the user devices

Server side:

- Web servers are the (software) server components of the network application running on a host machine

Both sides:

- Protocol shared by client and server components to communicate, i.e., HyperText Transfer Protocol (HTTP)

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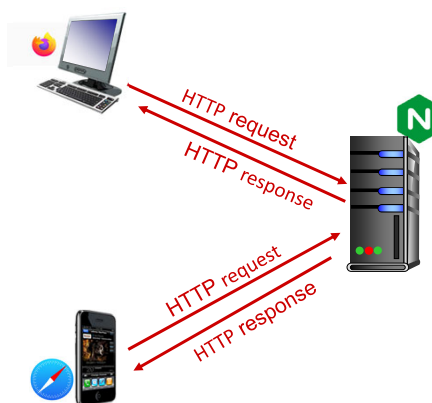
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Interactions

Client: the user agent requests, receives and “displays” web objects

Server: the web server sends web objects (or errors) in response to requests

Both client and server interact using the HTTP protocol



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Web history

In 1989 Sir Tim Berners-Lee at CERN (Switzerland) conceived the web as a tool for automatic information sharing among scientists spread around the world (Hypermedia information retrieval initiative)



He developed a web server and a textual browser

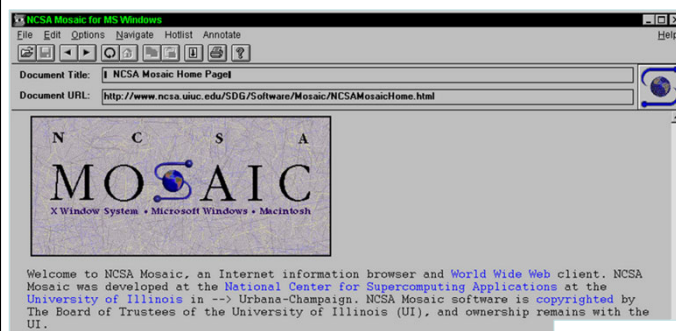


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In 1993 the first modern browser was developed by a group of students at NCSA for Unix platforms and it was named Mosaic



The New York Times

BUSINESS TECHNOLOGY

BUSINESS TECHNOLOGY; A Free and Simple Computer Link

Share full article

By John Markoff
Dec. 8, 1993

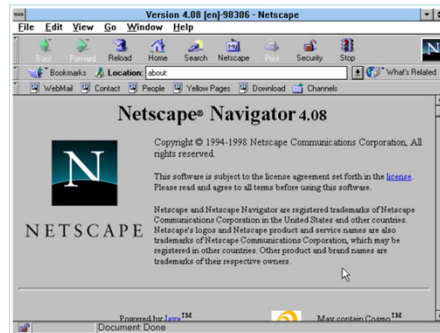
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Browser evolution

The first commercial browser (Netscape Navigator) with many advanced features (e.g., Javascript, https) was developed in 1995 by Netscape



In 1998 the Mozilla (Mosaic & Godzilla) project started from an open-source version of Netscape browser

In 2002 Mozilla Firefox (now at version 149.02) was released

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Other browsers

Opera, a browser targeting computers with limited resources, was released in 1994

Microsoft Internet Explorer was released in 1995 as part of the MS Windows 95 operating system (replaced by MS Edge since 2015)

In 2003 Apple Safari was released and in 2008 it was the turn of Google Chrome & Chromium project

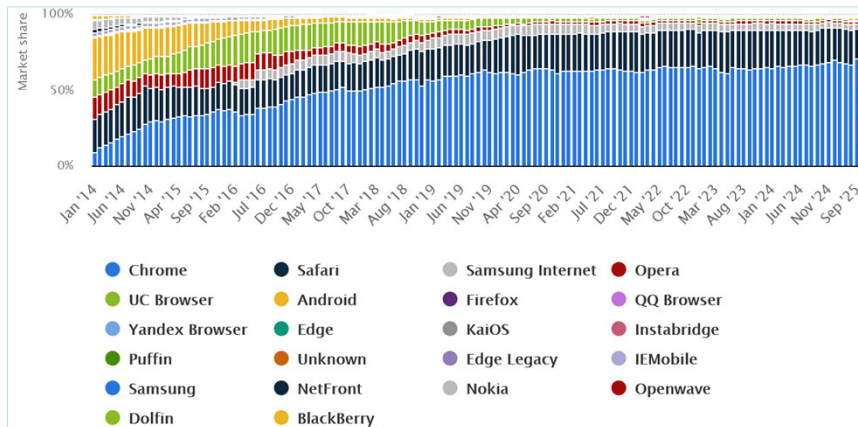
Other browsers (e.g., Brave, Vivaldi) released more recently

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Browser market share



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In September 2025, Google Chrome was the leader in mobile Internet browser market (71% share), followed by Apple's Safari (19.5% share)

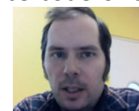
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Web server evolution

In 1993 Rob McCool at NCSA developed and released the source code of a simple `httpd`



After he left NCSA in 1994, a group of webmasters joined their efforts to patch and continue code development

In April 1995 Apache (A Patchy) server 0.8 was launched and in Dec. 1995 Apache 1.0 was released [the latest version is Apache 2.4.66 of Dec. 2025]

In 1995 Netscape Enterprise Server was released and also pre-release of Microsoft Internet Information Services

Other web servers:

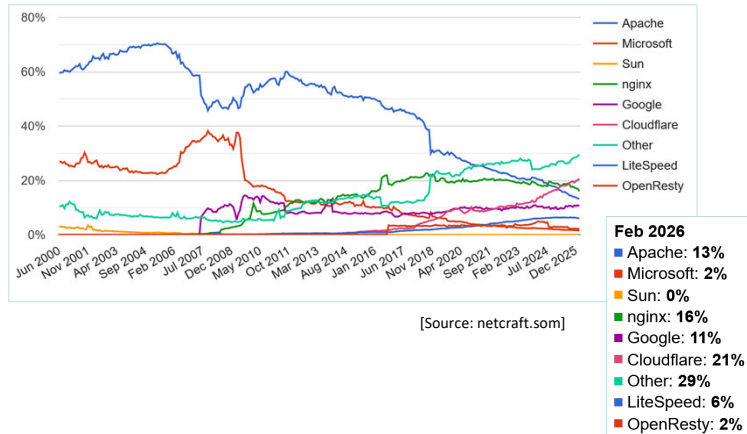
- nginx developed since 2001 for the Russian market
- Tengine created by Taobao for the Chinese market
- Google Web Server (GWS)
- Cloudflare server

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Web server market share



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HTTP overview

HTTP is a family of application-layer, open, stateless, request/response protocols for distributed, collaborative, hypertext information systems

Three major versions, i.e., HTTP/1.1, HTTP/2 and HTTP/3, each has specific benefits and limitations depending on the context

All versions share interface, semantics and messages [RFC 9110 “HTTP Semantics”, June 2022]

HTTP hides the details of how a service is implemented by presenting a uniform interface to clients independent of the types of resources

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HTTP overview (cont'd)

Basic unit of communication between clients and servers is the message

- Request messages sent by the clients
- Response messages sent by the servers

Data integrity requirements (e.g., TCP transport layer protocol)

Confidentiality is implemented using TLS-like features (**https**)

HTTP is a stateless protocol: servers consider requests in isolation rather than being associated with a specific client or a sequence of application steps

The relationships between messages do not have any impact on the interpretation of the request messages

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Stateless vs stateful protocols

Stateful protocols:

- Complex
- State (i.e., past history) must be maintained
- In case of a server or client crash, the views of the “state” might become inconsistent and must be reconciled

Stateless protocols:

- Simple
- No need to maintain any information about the past history
- Each request of a given client is “independent” of the others sent by a given client or by different clients
- No concept of session

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